

In its current revision, this project is an extension of Kurt et. al's 'LNmesh' implementation of offline Lightning Network payments using BLE-star and Wi-Fi [1]. LNmesh assumes that the payment channels for an offline transaction are already set up before the nodes go offline. Our project addresses this with a mesh node that has reliable internet access, which would facilitate channel opening and closing without requiring any direct internet access or prior channel setup. This raises concern about the security and integrity of transactions, since channels must be monitored during closing/dispute in order to detect any attempts to cheat by maliciously publishing outdated or dishonest channel states.

Papers [2] and [3] directly address the issue of monitoring channels to deter cheating while offline, and both provide valuable insight that prove useful. The authors of paper [2] address the issue of cheating with timelock attacks by proposing what they call CRAB, or Channel Resistant Against Bribery. This approach uses collateral during channel creation in order to incentivize the miners to publish punishment transactions rather than accept bribes for a timelock attack. This is useful for our project because it has the potential to reduce the reliance on watchtowers and reliable internet connection. Paper [3] takes a different approach for the same problem. Its authors designed a bi-directional payment channel using adaptor signatures, absolute timelocks, and a scheduled verification period. Despite those differences, it still addresses the idea of reducing the amount of monitoring required for LN channels.

In Paper [7], Bahrami et al. propose what they call LioTning, which is a custom, peer-supervised payment channel protocol for secure IoT payments. While it does not directly implement LN, it does still address the problem of channel supervision and proposes a new method to reduce reliance on a central monitoring party. This makes it relevant to our project because mesh-connected nodes may also benefit from distributed or peer-assisted monitoring when direct blockchain access is limited.

Beyond payment-channel monitoring, offline payment systems must also address transaction validity and double-spending when users cannot immediately verify transactions against a global ledger. Papers [6] and [8] examine this broader offline-payment security problem. A. M. Aghapour et al. [6] present "an offline method to prevent double spending with resource constrained mobile devices" [6]. Their method involves sharing information between payment devices about what funds have been spent, which may provide insight to whether certain funds have already been spent. This is useful because the authors report that they were able to approach the effectiveness of an ideal system using only about 240 bytes of metadata. In paper [8], H. Si et al. submit an asset-credential model using elliptic curve cryptography and zero knowledge proofs as a replacement for LN channels. Their work is useful because it addresses offline payment security and delayed verification, although it does not directly implement Lightning or mesh networking.

Paper [4] has similar motivations to ours in that the authors are also trying to implement offline payments, but their technical approach is different. They used Ethereum-compatible signed vouchers with a backend voucher coordinator to handle offline transactions, as opposed to bitcoin and a mesh network. They actually use BLE and QR codes for phone-to-phone

payment exchange. The main utility of this paper is in the payment coordination, and not necessarily in its implementation. Roberts et al. [5] also address offline payments in ad-hoc mobile environments. Like [4], this work is motivated by the need to support secure payments when traditional Internet infrastructure is unavailable. However, it differs from our project because it focuses on a broader offline mobile payment architecture rather than Bitcoin Lightning channels over a mesh network. Together, papers [4] and [5] are useful because they show different approaches to coordinating payment exchange in disconnected environments, even though they do not directly address Lightning channel opening or closing.

Finally paper [9] provides background knowledge about using mesh networking as decentralized communication environments. It is relevant because it discusses coordination and communication throughout mesh networks, but its utility for our purposes is limited because it does not discuss payment of any kind.

In order of relevance:

- [1] - Baseline
- [2], [3] - Monitoring LN channels for cheating
- [7] - Monitoring non-LN channels for cheating
- [6], [8] - Non-channel payment monitoring
- [4], [5] - Offline payment architecture
- [9] - Mesh architecture

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